

substitution0 

 

Medium Cryptography picoCTF 2022 Substitution

AUTHOR: WILL HONG

Description

A message has come in but it seems to be all scrambled. Luckily it seems to have the key at the beginning. Can you crack this substitution cipher?

Download the message [here](#).

Hints 

1

Try a frequency attack. An online tool might help.

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This challenge is a medium-level cryptography task from picoCTF 2022 that focuses on a Monoalphabetic Substitution Cipher.

Description

The challenge provides a scrambled message and mentions that the "key" is located at the beginning of the file.

- Provided Key: **DECKFMYIQJRWTPXGNABUSOLVH**
- Encrypted Message Snippet: **Ifnfuxpz Wfyndzk dnpaf, ogbi d yndsf dzk abdbfwv dan, dzk enpuyib tf bif effbwf...**

DECKFMYIQJRWTPXGNABUSOLVH

Ifnfuxpz Wfyndzk dnpaf, oqbi d yndsf dzk abdbfwv dan, dzk enpuyib tf bif effbwf mnpt d ywdaa cdaf qz oiqli qb oda fzcwpafk. Qb oda d efdubqmuw acdndedfua, dzk, db bidb bgtf, uzrzpoz bp zdbundwgaba-pm cpunaf d ynfdb xngfh qz d acqfzbqmqc xpqzb pm sqfo. Bifnf ofnf bop npuzk ewdcr axpba zfdn pzf flbnftqby pm bif edcr, dzk d wpzy pzf zfdn bif pbifn. Bif acdwfa ofnf flcffkqzywv idnk dzk ywpaav, oqbi dwv bif dxxfdndzcf pm eunzqaifk ypwk. Bif ofqyib pm bif qzafcb oda sfnv nftdnrdewf, dzk, bdrqzy dwv biqzya qzbp cpzaqkfndbapz, Q cpuwk idnkwy ewdtf luxqbfm mpn iqa pxqzqz nfaxfcbqzy qb.

Bif mwdy qa: xqcpCBM{5UE5717U710Z_3S0WU710Z_59533D2F}

The hints suggest using a frequency attack or online tools to crack the cipher.

Solution Strategy

A monoalphabetic substitution cipher replaces each letter of the alphabet with another fixed letter from a shuffled alphabet (the key).

Steps to solve:

1. **Identify the Substitution Alphabet:** The string **DECKFMYIQJRWTPXGNABUSOLVH** represents the mapping for the letters A-Z in order.
 - 'A' maps to 'D'
 - 'B' maps to 'E'
 - 'C' maps to 'C'
 - ...and so on.
2. **Use a Decryption Tool:** Tools like **dcode.fr** allow you to input the ciphertext and the specific substitution alphabet to automate the reversal.
3. **Manual Decryption (Optional):** Since the text follows standard English patterns (e.g., "Bif" likely maps to "The"), you can also solve it by mapping the provided key back to the standard alphabet.

dcode.fr/monoalphabetic-substitution



Search for a tool

★ [SEARCH A TOOL ON DCODE](#)
 e.g. type 'random'

★ [BROWSE THE FULL DCODE TOOLS' LIST](#)

Results

Alphabet : DECKFMYIQJRWTPXGNABUSOLVH
 Alphabet⁻¹ : STCABEQZHJDJXFRWOIKVMUYLPNG

Hereupon Legrand arose, with a grave and stately air, and brought me the beetle from a glass case in which it was enclosed. It was a beautiful scarabaeus, and, at that time, unknown to naturalists--of course a great prize in a scientific point of view. There were two round black spots near one extremity of the back, and a long one near the other. The scales were exceedingly hard and glossy, with all the appearance of burnished gold. The weight of the insect was very remarkable, and, taking all things into consideration, I could hardly blame Jupiter for his opinion respecting it.

The flag is:
picoCTF{5UB5717U710N_3V0LU710N_59533A2E}

Mono-alphabetic Substitution - [dCode](#)
 Tag(s) : Substitution Cipher

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MONO-ALPHABETIC SUBSTITUTION

Cryptography > Substitution Cipher > Mono-alphabetic Substitution

MONOALPHABETIC SUBSTITUTION DECODER

★ ALPHABETIC SUBSTITUTION CIPHERTEXT

Ifnfuxpz wfyndzk dnpaf, oqbi d yndsf dzk abdbfwv dqn,
 dzk enpuyib tf bif effbwf
 mnpt d yvdaa cdař qz oigci qb oda fzcwpafk. Qb oda d
 erfubgmuv acdndedfua, dzk. qb
 bidd batf. uzrpoz bp zdbundwqaba-pm cpunaf d ynfdb

★ SPACES ARE RELEVANT AND MUST BE KEPT (ARISTOCRAT CIPHER)
☐ CAN BE IGNORED OR ARE MISSING (PATRISTOCRAT CIPHER)

★ PLAINTEXT LANGUAGE English

▶ DECRYPT AUTOMATICALLY

OTHER DECRYPTION METHODS

☒ KNOWING THE SUBSTITUTION ALPHABET
 DECKFMYIQJRWTPXGNABUSOLVH

☐ KNOWING THE CODED/RECIPROCAL ALPHABET ↕

☐ KNOWING THE KEYWORD USED TO GENERATE THE ALPHABET

☐ ATTACK WITH PROBABLE WORD/KNOWN PLAIN TEXT SUBSTITUTION

☐ MANUAL DECRYPTION (EMPTY VISUAL TOOL)

▶ DECRYPT

See also: [Word Desubstitution/Pattern](#) – [Frequency Analysis](#)

MONOALPHABETIC SUBSTITUTION ENCODER

★ UNSUBSTITUTED PLAIN TEXT

dCode is able to decipher an alphabetic substitution of a text by analyzing the frequencies of occurrence of the letters. The message must be as long as possible so that dCode can accurately find the substitution alphabet.

★ SUBSTITUTION ALPHABET QWERTYUIOPASDFGHJKLZXCVBNM

Summary

- ★ Monoalphabetic Substitution Decoder
- ★ Monoalphabetic Substitution Encoder
- ★ Custom Deranged Alphabet Generator
- ★ What is a (mono-)alphabetical substitution? (Definition)
- ★ How to encrypt using an alphabetical substitution?
- ★ What do K1, K2, K3, K4 mean?
- ★ How to decrypt using an alphabetical substitution?
- ★ How to recognize a mono alphabetical substituted text?
- ★ How to decipher a substitution without the alphabet?
- ★ What is the MCMC technique?
- ★ What are the variants of the substitution cipher?

Similar pages

- ★ Word Desubstitution/Pattern
- ★ Deranged Alphabet Generator
- ★ Caesar Cipher

Final Result

After applying the substitution key to the ciphertext, the message becomes readable English text, ending with the flag.

Flag:

picoCTF{5UB5717U710N_3V0LU710N_59517LAD}